

MVC-Formulas

Parametric equations of line segment joining (x_0, y_0, z_0) and (x_1, y_1, z_1) is

$$x = x_0 + t(x_1 - x_0), \quad y = y_0 + t(y_1 - y_0), \quad z = z_0 + t(z_1 - z_0), \quad 0 \leq t \leq 1$$

Derivative of $\mathbf{r}(t) = \langle x(t), y(t), z(t) \rangle$ is

$$\mathbf{r}'(t) = \langle x'(t), y'(t), z'(t) \rangle$$

The arc length of any curve C given by $\mathbf{r}(t)$ where $a \leq t \leq b$ is

$$L = \int_a^b |\mathbf{r}'(t)| dt$$

In case of plane curve given by $\mathbf{r}(t) = \langle x(t), y(t) \rangle$, $a \leq t \leq b$, then the formula takes the form

$$L = \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$$

In case of space curve given by $\mathbf{r}(t) = \langle x(t), y(t), z(t) \rangle$, $a \leq t \leq b$, then the formula takes the form

$$L = \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2} dt$$

In case of $y = f(x)$ we can write $\mathbf{r}(x) = \langle x, f(x) \rangle$ and

$$L = \int_a^b \sqrt{1 + [f'(x)]^2} dx$$

The arc length function s is defined as

$$s(t) = \int_a^t |r'(u)| du \Rightarrow ds = |r'(t)| dt$$

An equation of tangent plane to the surface $z = f(x, y)$ at the point $P(x_0, y_0, z_0)$ is

$$z - z_0 = f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0)$$

The differential of $z = f(x, y)$ is

$$dz = z_x dx + z_y dy$$

For small $dx = \Delta x$ and $dy = \Delta y$

$$\Delta z \approx dz$$

If $F(x, y) = 0$ defines y implicitly as a differentiable function of x and $F_y \neq 0$ then

$$\frac{dy}{dx} = -\frac{F_x}{F_y}$$

If $F(x, y, z) = 0$ defines z implicitly as a differentiable function of x and y and $F_z \neq 0$ then

$$\frac{\partial z}{\partial x} = -\frac{F_x}{F_z} \quad \text{and} \quad \frac{\partial z}{\partial y} = -\frac{F_y}{F_z}$$

The gradient of $f(x, y)$ is

$$\nabla f(x, y) = \left\langle \frac{\partial}{\partial x}, \frac{\partial}{\partial y} \right\rangle f(x, y) = \left\langle \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right\rangle = \langle f_x, f_y \rangle$$

Similarly the gradient of $f(x, y, z)$ is

$$\nabla f(x, y, z) = \langle f_x, f_y, f_z \rangle$$

∇f give us the direction in which f changes fastest. $|\nabla f|$ is the maximum rate of change.

The directional derivative of $f(x,y)$ in the direction of a unit vector $u = \langle a, b \rangle$ is

$$D_{\mathbf{u}}f(x,y) = \nabla f(x,y) \cdot \mathbf{u}$$

Similarly if $\mathbf{u} = \langle a, b, c \rangle$ is a unit vector and $f(x,y,z)$ is a three variables function then

$$D_{\mathbf{u}}f(x,y,z) = \nabla f(x,y,z) \cdot \mathbf{u}$$

If $f(x,y) \geq 0$, then the volume of the solid that lies above the rectangle R and below the surface $z = f(x,y)$ is

$$V = \iint_R f(x,y) dA$$

Fubini's Theorem: If f is continuous on the rectangle

$$R = \{(x,y) \mid a \leq x \leq b, c \leq y \leq d\}$$

then

$$\iint_R f(x,y) dA = \int_a^b \int_c^d f(x,y) dy dx = \int_c^d \int_a^b f(x,y) dx dy$$

$$\iint_R g(x)h(y) dA = \int_a^b g(x) dx \int_c^d h(y) dy \quad \text{where } R = [a,b] \times [c,d].$$

The average value of a function $f(x,y)$ over a region R is

$$f_{avg} = \frac{1}{A(R)} \iint_R f(x,y) dA$$

If f is continuous on a type I region described by

$$D = \{(x, y) \mid a \leq x \leq b, g_1(x) \leq y \leq g_2(x)\}$$

then

$$\iint_D f(x, y) dA = \int_a^b \int_{g_1(x)}^{g_2(x)} f(x, y) dy dx$$

If f is continuous on a type II region described by

$$D = \{(x, y) \mid c \leq y \leq d, h_1(y) \leq x \leq h_2(y)\}$$

then

$$\iint_D f(x, y) dA = \int_c^d \int_{h_1(y)}^{h_2(y)} f(x, y) dx dy$$

If $D = D_1 \cup D_2$ and D_1 and D_2 don't overlap then

$$\iint_D f(x, y) dA = \iint_{D_1} f(x, y) dA + \iint_{D_2} f(x, y) dA$$

$$\iint_D 1 dA = A(D)$$

Polar coordinates (r, θ) of a point are related to the rectangular coordinates (x, y) by the equations

$$x = r \cos \theta \quad y = r \sin \theta \quad x^2 + y^2 = r^2$$

Change to Polar Coordinates in a Double Integral: If f is continuous on a polar rectangle R given by $0 \leq a \leq r \leq b$, $\alpha \leq \theta \leq \beta$, where $0 \leq \beta - \alpha \leq 2\pi$,

then

$$\iint_R f(x,y) dA = \int_{\alpha}^{\beta} \int_a^b f(r \cos \theta, r \sin \theta) r dr d\theta$$

If a smooth surface S is given by

$$\mathbf{r}(u,v) = \langle u(x,y), y(u,v), z(u,v) \rangle \quad (u,v) \in D$$

and S is covered just once as (u,v) varies throughout D , then the area of S is

$$A(S) = \iint_D |\mathbf{r}_u \times \mathbf{r}_v| dA$$

In case of $z = f(x,y)$, $(x,y) \in D$ we can describe the surface by

$$\mathbf{r}(x,y) = \langle x, y, f(x,y) \rangle, \quad (x,y) \in D$$

and

$$A(S) = \iint_D |\mathbf{r}_x \times \mathbf{r}_y| dA = \iint_D \sqrt{z_x^2 + z_y^2 + 1} dA$$

Similarly in case of $y = f(x,z)$ we have

$$A(S) = \iint_D \sqrt{y_x^2 + y_z^2 + 1} dA$$

and in case of $x = f(y,z)$ we have

$$A(S) = \iint_D \sqrt{x_y^2 + x_z^2 + 1} dA$$

If $B = [a,b] \times [c,d] \times [r,s]$ then

$$\iiint_B f(x,y,z) dV = \int_a^b \int_c^d \int_r^s f(x,y,z) dz dy dx$$

We can evaluate above triple integral in five other ways.

If $E = \{(x, y, z) \mid (x, y) \in D, u_1(x, y) \leq z \leq u_2(x, y)\}$ then

$$\iiint_E f(x, y, z) dV = \iint_D \left[\int_{u_1(x, y)}^{u_2(x, y)} f(x, y, z) dz \right] dA$$

If $E = \{(x, y, z) \mid (x, z) \in D, u_1(x, z) \leq y \leq u_2(x, z)\}$ then

$$\iiint_E f(x, y, z) dV = \iint_D \left[\int_{u_1(x, z)}^{u_2(x, z)} f(x, y, z) dy \right] dA$$

If $E = \{(x, y, z) \mid (y, z) \in D, u_1(y, z) \leq x \leq u_2(y, z)\}$ then

$$\iiint_E f(x, y, z) dV = \iint_D \left[\int_{u_1(y, z)}^{u_2(y, z)} f(x, y, z) dx \right] dA$$

Parametric equations of sphere are

$$x = \rho \sin \phi \cos \theta, \quad y = \rho \sin \phi \sin \theta, \quad z = \rho \cos \phi, \quad 0 \leq \phi \leq \pi, 0 \leq \theta \leq 2\pi$$

Parametric equations of circle $(x - h)^2 + (y - k)^2 = r^2$ are

$$x = r \cos t + h, \quad y = r \sin t + k, \quad 0 \leq t \leq 2\pi$$

To parametrize $y = f(x)$ we can take $x = t$ and $y = f(t)$. Similarly to parametrize $x = h(y)$, we can take $y = t$ and $x = h(t)$.

A vector field on R^2 can be written as

$$F(x, y) = \langle P(x, y), Q(x, y) \rangle$$

A vector field on R^3 can be written as

$$F(x, y) = \langle P(x, y), Q(x, y), R(x, y) \rangle$$

If a plane curve C is given by $\mathbf{r}(t) = \langle x(t), y(t) \rangle$, $a \leq t \leq b$ then

$$\int_C f(x,y)ds = \int_a^b f(x(t),y(t))|\mathbf{r}'(t)|dt = \int_a^b f(\mathbf{r}(t))|\mathbf{r}'(t)|dt \quad (\text{In short})$$

Also if a space curve C is given by $\mathbf{r}(t) = \langle x(t), y(t), z(t) \rangle$, $a \leq t \leq b$ then

$$\int_C f(x,y,z)ds = \int_a^b f(x(t),y(t),z(t))|\mathbf{r}'(t)|dt = \int_a^b f(\mathbf{r}(t))|\mathbf{r}'(t)|dt \quad (\text{In short})$$

Remember: $ds = |\mathbf{r}'(t)|dt$

Similarly,

$$\int_C f(x,y)dx = \int_a^b f(\mathbf{r}(t))x'(t)dt$$

and

$$\int_C f(x,y)dy = \int_a^b f(\mathbf{r}(t))y'(t)dt$$

Remember: $dx = x'(t)dt$ and $dy = y'(t)dt$

Line integral of a vector field $\mathbf{F}$ along C is

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_a^b F(\mathbf{r}(t)) \cdot \mathbf{r}'(t)dt$$

Note: Work done by F in moving a particle along C is

$$W = \int_C \mathbf{F} \cdot d\mathbf{r} = \int_a^b F(\mathbf{r}(t)) \cdot \mathbf{r}'(t)dt$$

If $F = \langle P, Q, R \rangle$, then

$$W = \int_C \mathbf{F} \cdot d\mathbf{r} = \int_C Pdx + Qdy + Rdz$$

Remember: $d\mathbf{r} = \mathbf{r}'(t)dt$

$$\begin{array}{rcl}
\int_{-C} f(x,y) ds & = & - \int_C f(x,y) ds \\
\int_{-C} f(x,y) dx & = & \int_C f(x,y) dx \\
\int_{-C} f(x,y) dy & = & \int_C f(x,y) dy \\
\int_{-C} \mathbf{F} \cdot d\mathbf{r} & = & - \int_C \mathbf{F} \cdot d\mathbf{r}
\end{array}$$

If $\mathbf{F}$ is a vector field then

$$\text{curl}(\mathbf{F}) = \nabla \times \mathbf{F} \quad \text{and} \quad \text{div}(\mathbf{F}) = \nabla \cdot \mathbf{F}$$

If $\text{curl}(\mathbf{F}) = \mathbf{0}$ then $\mathbf{F}$ is called irrotational.

If $\text{div}(\mathbf{F}) = 0$ then $\mathbf{F}$ is called incompressible.

A vector field $\mathbf{F}$ is called a conservative vector field if it is the gradient of some scalar function f .

$$\mathbf{F} = \nabla f$$

In this situation f is called a potential function for $\mathbf{F}$.

Theorem: If $\mathbf{F}$ is continuous every where and $\text{curl}(\mathbf{F}) = \mathbf{0}$ then $\mathbf{F}$ is conservative.

If $\mathbf{F}(x,y) = \langle P(x,y), Q(x,y) \rangle$ then

$$\text{curl}(\mathbf{F}) = \nabla \times \mathbf{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & 0 \\ P & Q & 0 \end{vmatrix} = \langle 0, 0, Q_x - P_y \rangle$$

Thus $\mathbf{F}(x,y) = \langle P(x,y), Q(x,y) \rangle$ is conservative if $Q_x = P_y$.

Fundamental Theorem for line Integral is

$$\int_C \nabla f \cdot d\mathbf{r} = f(\mathbf{r}(b)) - f(\mathbf{r}(a))$$

If S is a surface given by

$$\mathbf{r}(u, v) = \langle x(u, v), y(u, v), z(u, v) \rangle, \quad (u, v) \in D$$

then

$$\iint_S f(x, y, z) dS = \iint_D f(\mathbf{r}(u, v)) |\mathbf{r}_u \times \mathbf{r}_v| dA$$

Remember: $dS = |\mathbf{r}_u \times \mathbf{r}_v| dA$

If $z = g(x, y)$, where $(x, y) \in D$ is the surface then we can write

$\mathbf{r}(x, y) = \langle x, y, g(x, y) \rangle$ and $|\mathbf{r}_x \times \mathbf{r}_y| = \sqrt{z_x^2 + z_y^2 + 1}$, thus

$$\iint_S f(x, y, z) dS = \iint_D f(x, y, g(x, y)) \sqrt{z_x^2 + z_y^2 + 1} dA$$

Similarly if $y = g(x, z)$, where $(x, z) \in D$ is the surface then

$$\iint_S f(x, y, z) dS = \iint_D f(x, g(x, z), z) \sqrt{y_x^2 + y_z^2 + 1} dA$$

and if $x = g(y, z)$, where $(y, z) \in D$ is the surface then

$$\iint_S f(x, y, z) dS = \iint_D f(g(y, z), y, z) \sqrt{x_y^2 + x_z^2 + 1} dA$$

Green's Theorem States

$$\int_C P dx + Q dy = \iint_D (Q_x - P_y) dA$$

where C is a positively oriented, piecewise-smooth, simple closed curve in the plane and D is the region enclosed by C .

The unit normal to a surface $\mathbf{r}(u, v)$ is

$$\mathbf{n} = \frac{\mathbf{r}_u \times \mathbf{r}_v}{|\mathbf{r}_u \times \mathbf{r}_v|}$$

The surface integral of a vector field $\mathbf{F}$ is

$$\iint_S \mathbf{F} \cdot d\mathbf{S} = \iint_S \mathbf{F} \cdot \mathbf{n} dS = \iint_D \mathbf{F}(\mathbf{r}(u, v)) \cdot |\mathbf{r}_u \times \mathbf{r}_v| dA$$

where D is the parameter domain (which is some region in the uv -plane). Note that the above integral give us FLUX of $\mathbf{F}$ across S , which in simple words, is the rate of flow through the surface.

Remember: $d\mathbf{S} = \mathbf{n}dS = (\mathbf{r}_u \times \mathbf{r}_v)dA$

Stokes' Theorem states

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \iint_S \text{curl}(\mathbf{F}) \cdot d\mathbf{S}$$

where S is an oriented smooth surface and C is a simple, closed, piecewise-smooth boundary curve of S .

Divergence Theorem states

$$\iint_S \mathbf{F} \cdot d\mathbf{S} = \iiint_E \text{div}\mathbf{F} dV$$

where E is a solid region and S is the boundary surface of E .

❶ $\int_a^b 1 dx = \text{Length of } [a, b]$

❷ $\iint_D 1 dA = A(D)$

❸ $\iiint_E 1 dV = V(E)$

❹ $\int_C 1 ds = L$, which is arc length

$$\textcircled{5} \quad \iint_S 1 dS = A(S)$$

The End